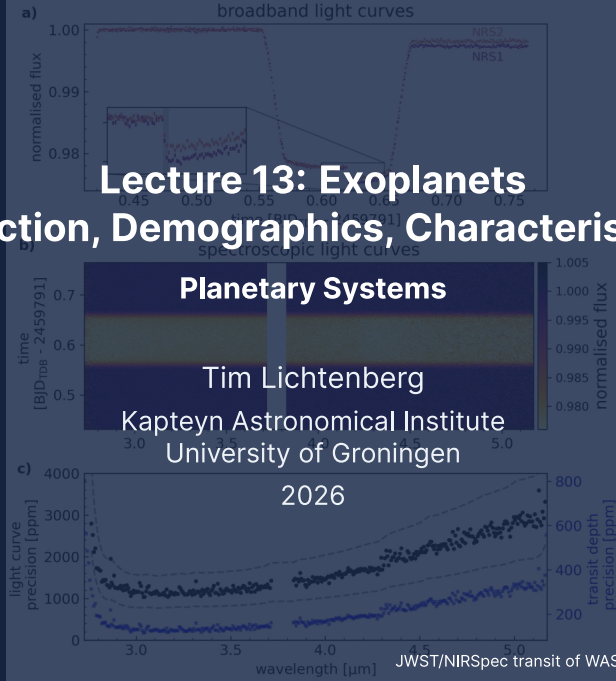


Lecture 13: Exoplanets

Detection, Demographics, Characterisation



Learning Objectives

By the end of this lecture, you will be able to:

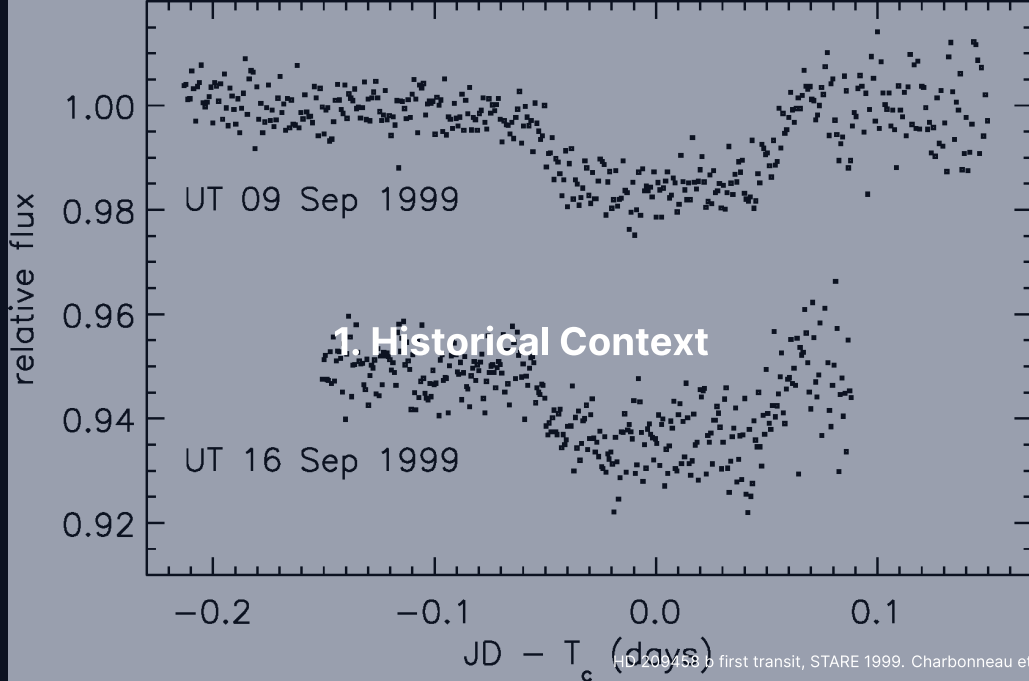
- Describe the main exoplanet detection methods and their biases
- Interpret the period-radius diagram and the radius valley
- Apply transit and radial-velocity geometry to derive bulk density
- Evaluate JWST-era atmospheric results and biosignature claims

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

- Lectures 9–11 surveyed the geology, atmospheres, and interiors of the solar planets
- Lecture 12 decoded small bodies and meteorites as planet formation fossils
- This lecture turns outward to thousands of exoplanets to test if these patterns are universal

Today: How do we detect, weigh, and characterise worlds beyond our solar system?



1. Historical Context

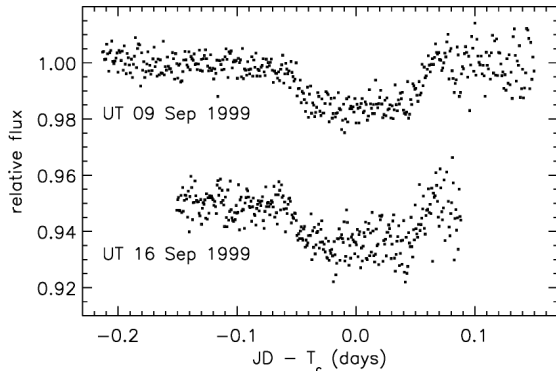
HD 209458 b first transit, STARE 1999. Charbonneau et al. (2000)

From Zero to Six Thousand in Three Decades

Three decades: from the first detection to a census of over 6000 confirmed worlds.

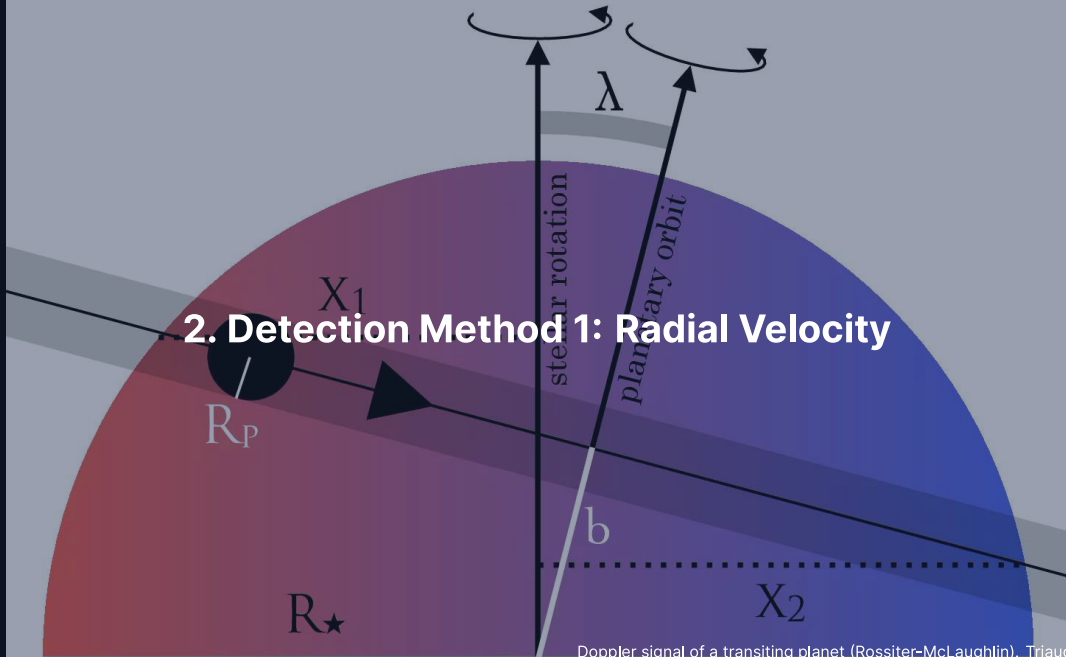
- **1992 (PSR B1257+12):** first exoplanets, around a pulsar, by timing (Wolszczan & Frail)
- **1995 (51 Pegasi b):** first planet around a Sun-like star; a 4.23-day orbit forced migration theory (Mayor & Queloz)
- **1999 (HD 209458 b):** first transit of an RV planet: a gas-giant radius and bulk density

HD 209458 b: First Ground-Based Transit (1999)



- STARE photometer, two nights in Sep 1999
- Successive transits dropped flux by ~ 1.7%
- Independent detection by Henry et al. (2000)
- Exoplanets became physical objects with measurable radii

2. Detection Method 1: Radial Velocity

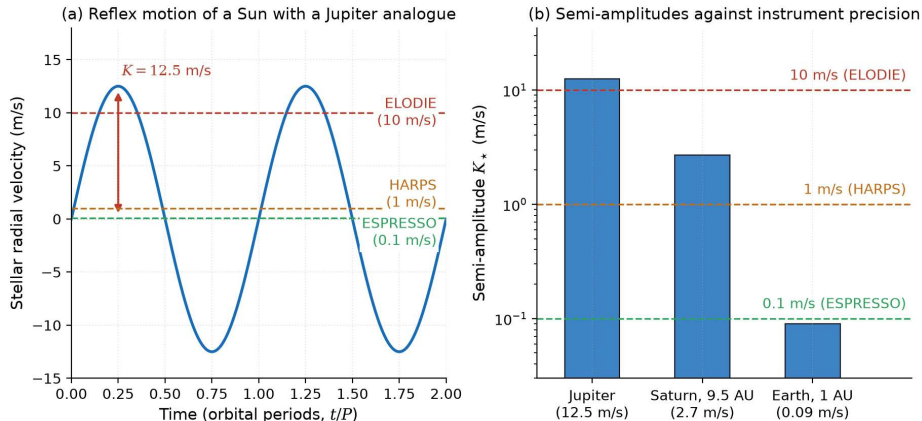


Radial Velocity: Doppler Reflex of the Star

- Star and planet orbit a common barycentre: $a_{\star} = (m_p/M_{\star}) a_p$
- Periodic Doppler shift of stellar absorption lines yields reflex velocity
- Semi-amplitude $K_{\star}$ (edge-on orbit; set $e = 0$ for a circular one):

$$K_{\star} = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

- Jupiter at 5.2 AU: $K_{\star} \approx 12.5 \text{ m s}^{-1}$
- Earth at 1 AU: $K_{\star} \approx 9 \text{ cm s}^{-1}$

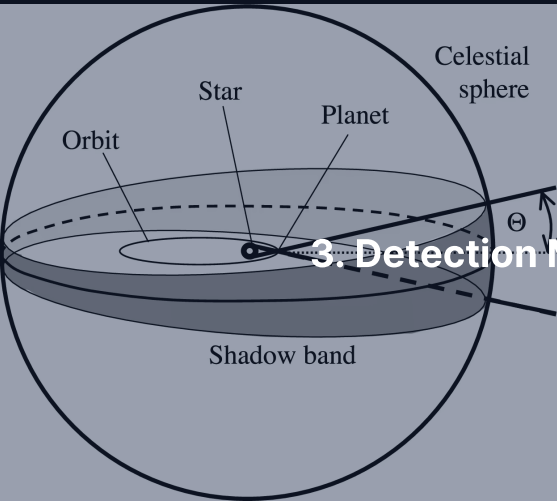


(a) The reflex velocity of a Sun with a Jupiter analogue, $K_* = 12.5 \text{ m s}^{-1}$, against the precisions of ELODIE (10 m s^{-1}), HARPS (1 m s^{-1}) and ESPRESSO (0.1 m s^{-1}). (b) The semi-amplitudes of a Jupiter analogue, Saturn at 9.5 AU (2.7 m s^{-1}) and Earth at 1 AU (0.09 m s^{-1}): an Earth analogue sits below the instrumental floor and within the stellar noise. Course figure.

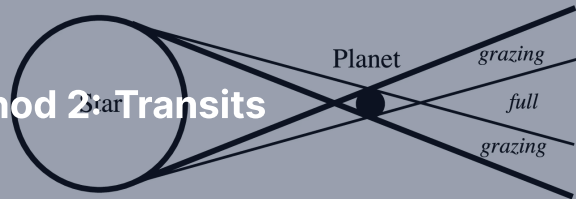
Noise Floor and the $m \sin i$ Degeneracy

RV measures only minimum mass $m_p \sin i$, with precision now limited by stellar jitter rather than spectrographs.

- **Noise floor:** ESPRESSO reaches 10 cm s^{-1} , but stellar granulation and spots produce $0.1\text{--}1 \text{ m s}^{-1}$ jitter
- **Degeneracy:** Orbit inclination i is unknown, so RV alone yields only a lower limit $m_p \sin i$
- **Breaking degeneracy:** Transits guarantee $\sin i \approx 1$, collapsing the ambiguity to yield the true planet mass

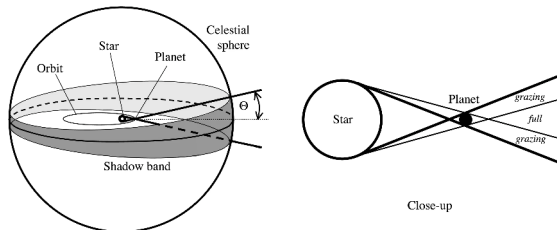


3. Detection Method 2: Transits



Close-up

Transit Geometry



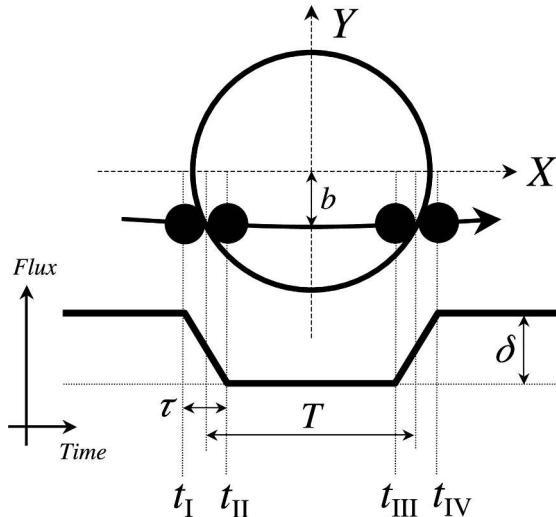
- Orbital plane near line of sight
→ planet crosses stellar disk
- Transit depth: $\delta = (R_p/R_\star)^2$
- Geometric probability:
 $p \approx R_\star/a$
- Strong selection bias toward short periods and large $R_p/R_\star$

Quick Scales

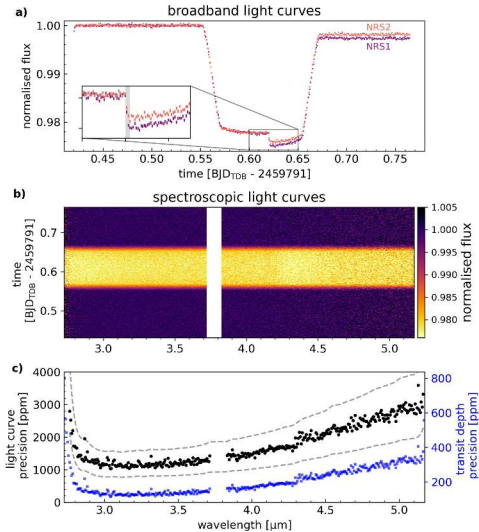
Transit depth scales inversely with the square of stellar radius, making small planets around small stars far easier to detect.

- **Jupiter around Sun:** $\delta \approx 1.01\%$, easily detected from ground-based telescopes
- **Earth around Sun:** $\delta \approx 84$ ppm, requires ultra-precise space photometry
- **Earth around M dwarf ($0.15 R_{\odot}$):** $\delta \approx 4000$ ppm, accessible from ground and space

Light Curve Anatomy

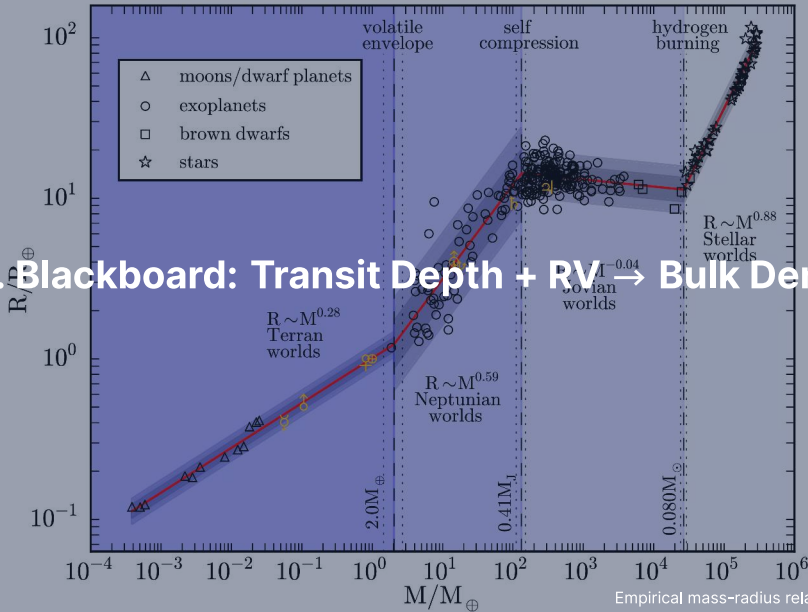


- Four contact times: t_I , t_{II} , t_{III} , t_{IV}
- Duration yields impact parameter b and orbital inclination
- Ingress and egress shape constrains stellar limb darkening
- Transit depth determines planet-to-star radius ratio $R_p/R_\star$



JWST/NIRSpec transit photometry of WASP-39 b: transit depth measured to 200–600 ppm per spectroscopic bin, pushing photometric noise below typical exoplanet absorption features. Alderson et al. (2023).

4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



Why Combine Transit and RV?

- Transit alone: yields planet radius R_p , but no mass
- RV alone: yields minimum mass $m_p \sin i$, but no radius
- Combined: **both** for the same planet $\rightarrow$ bulk density $\bar{\rho}_p$
- Bulk density turns an astronomical detection into a physical, geological body

To the board.

Result: Depth, Semi-Amplitude, Density

Transit depth $\delta = (R_p/R_\star)^2$ gives R_p ; the reflex velocity

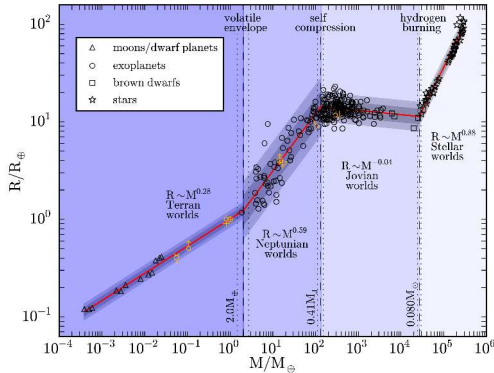
$K_\star = (2\pi G/P)^{1/3} m_p \sin i M_\star^{-2/3}$ gives $m_p \sin i$ (broad).

- A transit forces $\sin i \approx 1$ and collapses the $m_p \sin i$ degeneracy
- Both R_p and m_p measured for the same object
- Bulk density follows from elementary geometry:

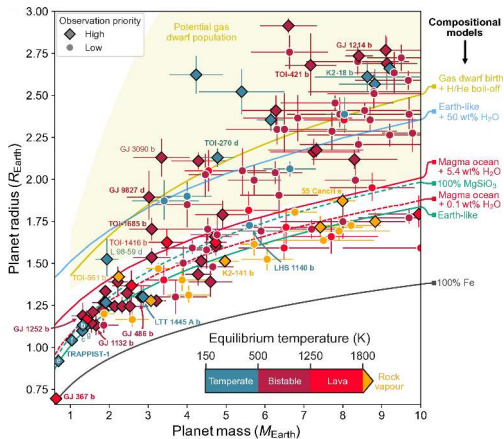
$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.

Density Maps to Composition

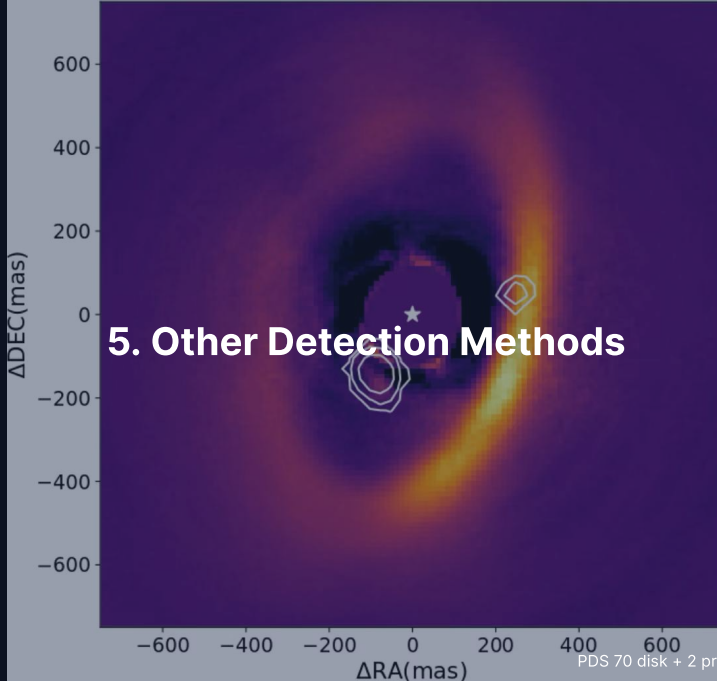


- $\bar{\rho} \approx 5.5 \text{ g cm}^{-3}$: rocky terrestrial (Earth, Venus)
- $\bar{\rho} \approx 1.3 \text{ g cm}^{-3}$: gas giant (Jupiter)
- $\bar{\rho} \approx 0.5 \text{ g cm}^{-3}$: inflated volatile envelope
- $\bar{\rho} \approx 2\text{--}4 \text{ g cm}^{-3}$: sub-Neptunes, unseen in the solar system



Mass against radius for small exoplanets with interior models from pure iron to volatile-rich envelopes, sorted by thermal regime: rocky super-Earths follow the silicate-iron curves, sub-Neptunes need substantial volatile envelopes or water-rich layers to explain their larger radii. Reproduced from Lichtenberg et al. (2025).

5. Other Detection Methods

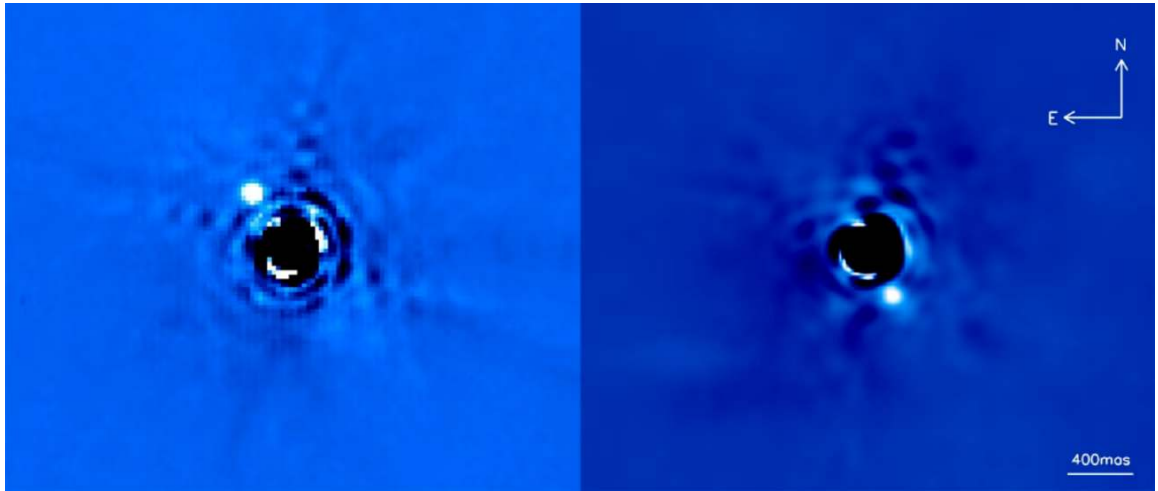


PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

Direct Imaging: Spatial Separation of Photons

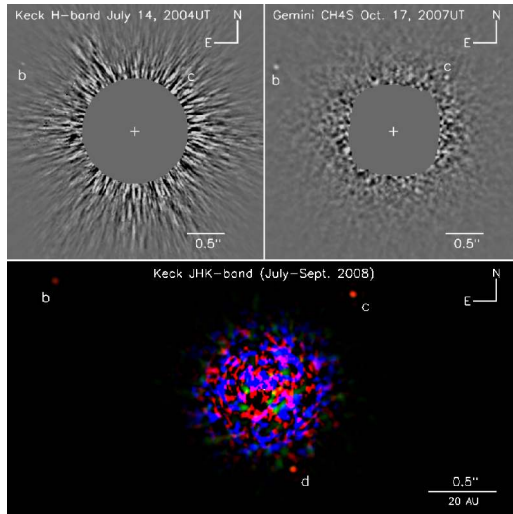
Direct imaging separates planet photons from the stellar glare.

- **Contrast:** suppress the star by 10^{-9} (Jupiter) to 10^{-10} (Earth twin) at separation $\theta = a/d$
- **Technology:** extreme adaptive optics plus coronagraphs against diffraction and speckle noise
- **Bias:** young, massive, self-luminous giants on wide orbits ($a > 10$ AU)

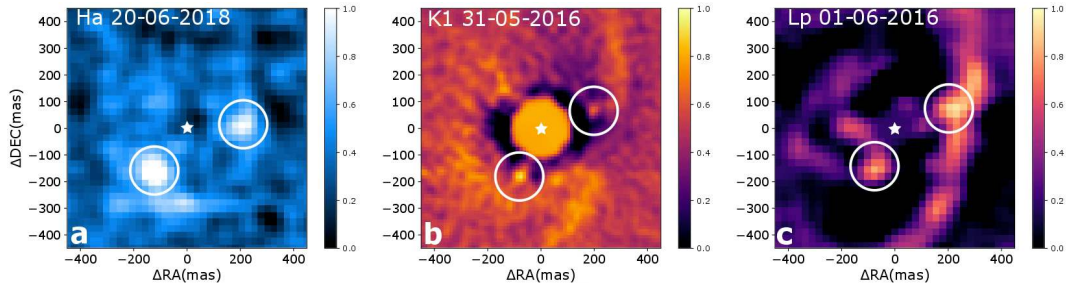


Direct image of β Pictoris b from Lagrange et al. (2010), showing orbital motion between 2003 and 2009 that confirms a bound $\sim 9 M_J$ companion at ~ 9 AU. Credit: ESO/A.-M. Lagrange et al.

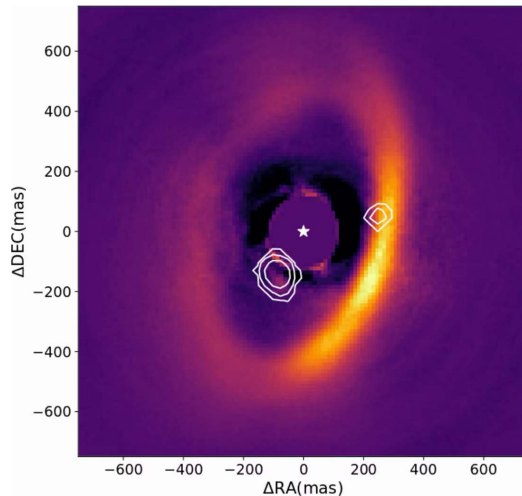
HR 8799: Four Imaged Giants



- Keck and Gemini AO discovery of three bound co-moving companions (Marois et al. 2008); planet e added in 2010
- Four planets confirmed across multiple epochs
- Masses $5\text{--}10 M_J$ at orbital separations 14–70 AU
- Young host star (30–60 Myr): the planets still glow from their contraction heat



PDS 70 b and c inside the disk gap in three bands: MUSE H α (a), SPHERE K1 (b), NACO L' (c); both protoplanets are actively accreting within the cleared cavity. Haffert et al. (2019).



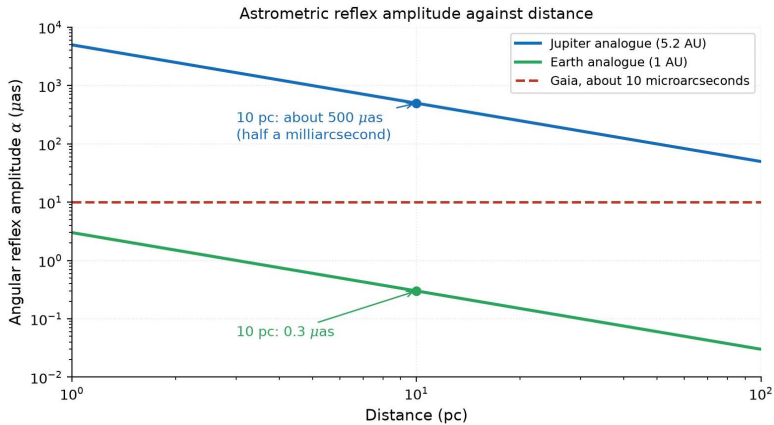
Composite scattered-light image of the PDS 70 protoplanetary disk showing the parent circumstellar disk and two embedded protoplanets inside the cleared gap. Credit: Haffert et al. (2019).

Astrometry: Gaia Will Find Jupiter Analogues

Astrometry follows the star's reflex wobble on the sky and gives true masses.

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- **Signal:** a Jupiter analogue at 10 pc gives $\alpha \sim 500 \mu\text{as}$; an Earth analogue $\sim 0.3 \mu\text{as}$
- **Gaia DR4 (2026):** thousands of cold giants expected; $\sim 10 \mu\text{as}$ final precision
- **True mass:** the inclination i is measured, so no $m \sin i$ degeneracy



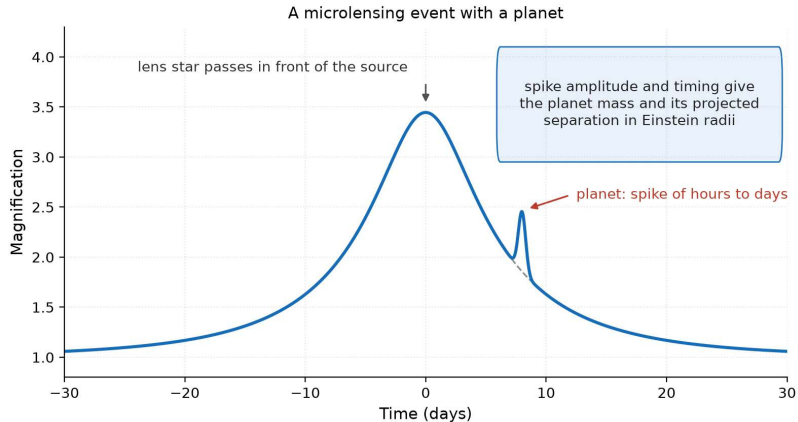
Astrometric reflex amplitude $\alpha = (m_p / M_*) (a_p / d)$ against distance for a Jupiter analogue at 5.2 AU and an Earth analogue at 1 AU: at 10 pc about 500 microarcseconds (half a milliarcsecond) and 0.3 microarcseconds. The dashed line is Gaia's final precision of about 10 microarcseconds for bright stars, the regime in which Jupiter analogues become accessible.

Course figure.

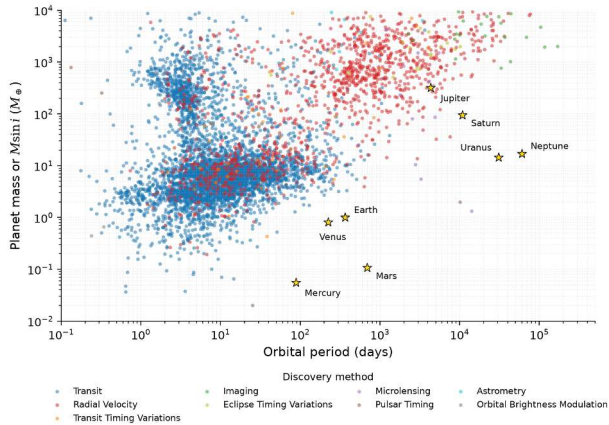
Microlensing, Timing, and Detection Biases

Each method covers its own part of parameter space, so the archive carries the biases of the methods.

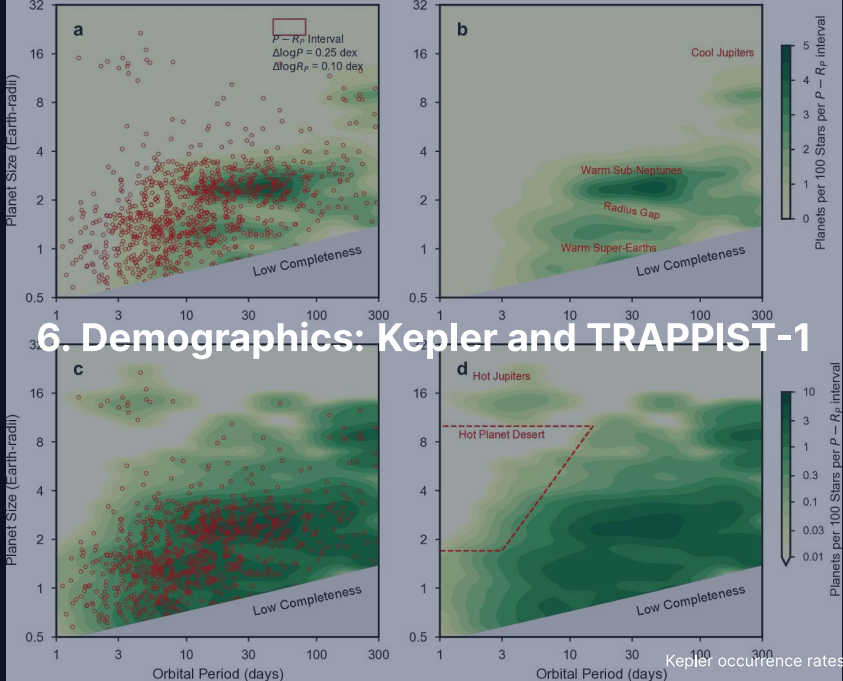
- **Microlensing (Roman):** cool planets at 1–10 AU and unbound rogue worlds
- **Timing variations (TTVs):** dynamical masses for faint hosts such as TRAPPIST-1
- **Together:** transits and RV probe close-in orbits; imaging, astrometry and microlensing the wide ones



A microlensing event with a planet: the magnification of a background source as the lens star passes in front of it (impact parameter 0.3 Einstein radii, Einstein crossing time 15 days), and the spike of hours to days that a planet around the lens adds; its amplitude and timing give the planet's mass and projected separation in Einstein radii. Course figure.



Confirmed exoplanet masses (or minimum masses $m_p \sin i$) against orbital period, coloured by detection technique, with the solar-system planets for comparison: detection biases concentrate discoveries at high mass and short period and leave Earth analogues in a sparsely sampled region. Credit: NASA Exoplanet Archive cite:p'NASAExoplanetArchive2026'. Course figure.

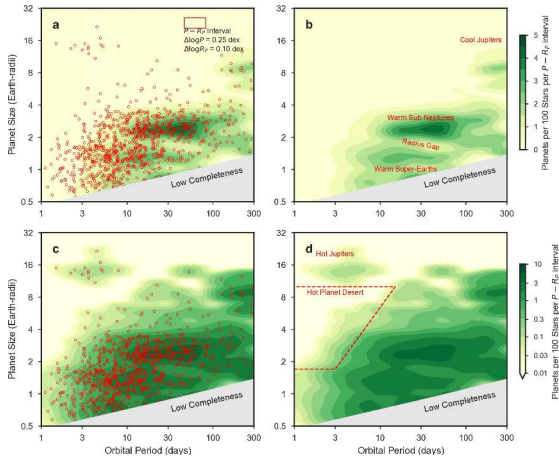


The Kepler Revolution

Kepler's computable detection efficiency delivered the first bias-corrected occurrence rates: planets outnumber stars in the Milky Way.

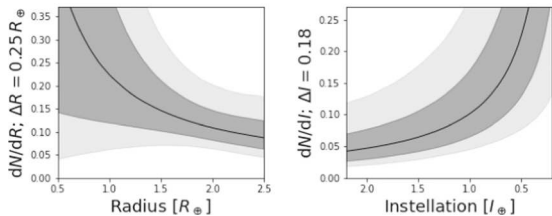
- **Planets are ubiquitous:** Average of > 1 planet per main-sequence star (Petigura et al. 2018)
- **Small planets dominate:** Super-Earths and sub-Neptunes are common; hot Jupiters are rare (0.5–1%)
- **Habitable terrestrial rate:** Occurrence of Earth-sized planets in the habitable zone is $\eta_{\oplus} \sim 0.4$ (Bryson et al. 2021)

Period-Size Occurrence Rate



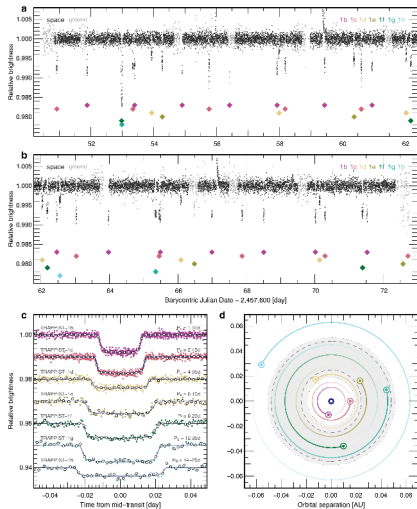
- California-Kepler Survey, after stellar parameter refinement
- Small planets common across all orbital periods
- Small planets outnumber giants at every period
- Fall-off beyond ~ 100 days is partly completeness-limited

$\eta_{\oplus}$: Earths in Habitable Zones



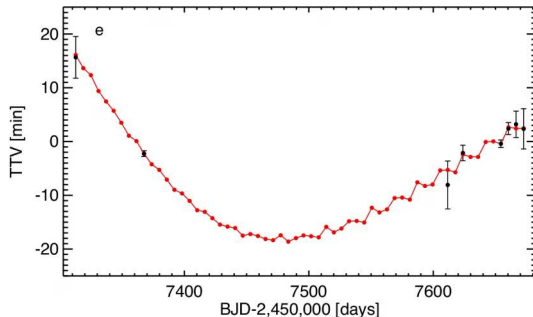
- Bryson et al. (2021) occurrence rate for Sun-like stars
- Marginalised over planet radius and stellar instellation
- Shaded bands represent 68% and 95% credible intervals
- Central estimate $\eta_{\oplus} \sim 0.4$: terrestrial planets in habitable zones are common

TRAPPIST-1: Seven Earths in Resonance

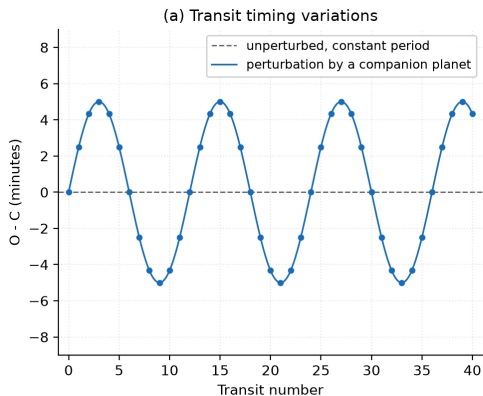


- M8 ultra-cool dwarf at 12 pc hosting seven Earth-sized worlds within 0.06 AU
- Resonant Laplace-like chain is consistent with early inward disk migration (Gillon et al. 2017)
- Three planets (e, f, g) orbit in the temperate habitable zone (Gillon et al. 2017)

TRAPPIST-1 Masses from TTVs



- Multi-planet transit-timing variations reach tens-of-minutes amplitude
- Simultaneous N-body dynamical modeling breaks mass degeneracies
- Determines individual masses and densities without requiring radial velocities



(b) Timing methods

transit timing variations:

dynamical masses (TRAPPIST-1)

pulsar timing:

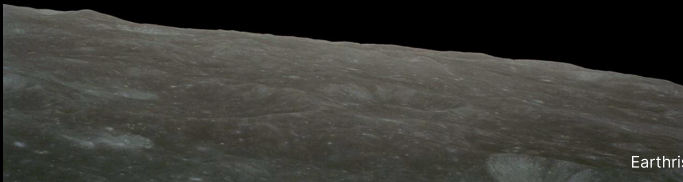
the first exoplanets (1992)

eclipse timing in binaries:

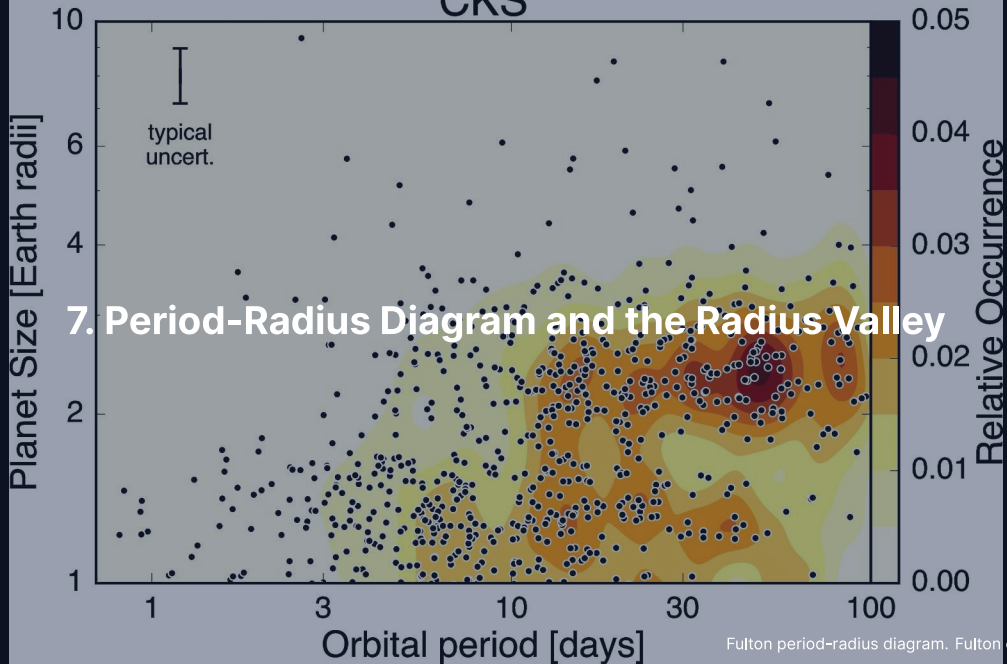
circumbinary planets (Kepler-16 b, 2011)

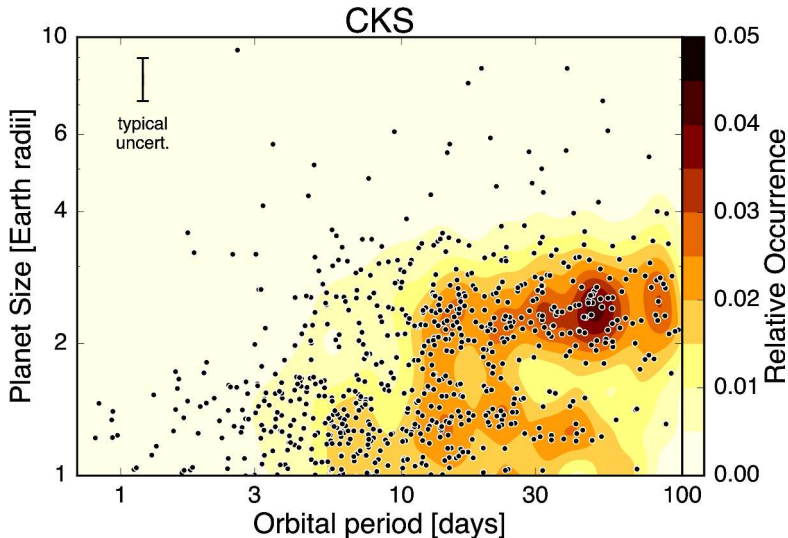
(a) Transit timing variations: the observed minus calculated transit times of a planet perturbed by a companion oscillate about the constant-period prediction (5 minute amplitude over 12 transits, schematic); amplitude and period encode the companion's mass. (b) The three timing methods: transit timing variations (dynamical masses, TRAPPIST-1), pulsar timing (the first exoplanets, 1992) and eclipse timing in binaries (circumbinary planets, Kepler-16 b in 2011). Course figure.

Break



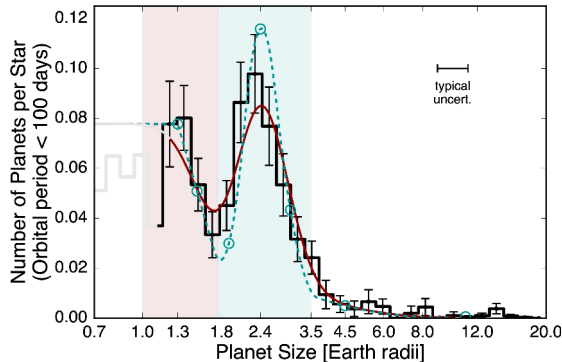
CKS





Period-radius distribution of small Kepler planets after stellar parameter refinement, revealing super-Earth and sub-Neptune populations separated by a deficit at $\sim 1.8 R_{\oplus}$. Fulton et al. (2017).

The Radius Valley (Fulton Gap)



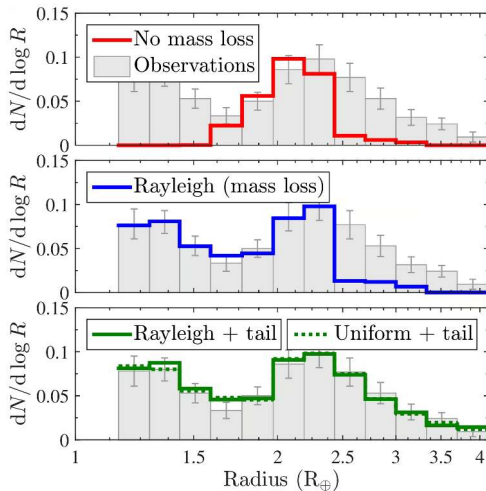
- Statistically significant deficit at $1.5\text{--}2.0 R_{\oplus}$
- Super-Earth population peaks at $\sim 1.3 R_{\oplus}$ (bare rocky cores)
- Sub-Neptune population peaks at $\sim 2.4 R_{\oplus}$ (volatile envelopes)
- Two stripping mechanisms: photoevaporation (stellar XUV) and core-powered mass loss (interior heat, no XUV needed)

Photoevaporation Strips H/He Envelopes

Energy-limited hydrodynamic escape driven by high stellar X-ray and EUV irradiation:

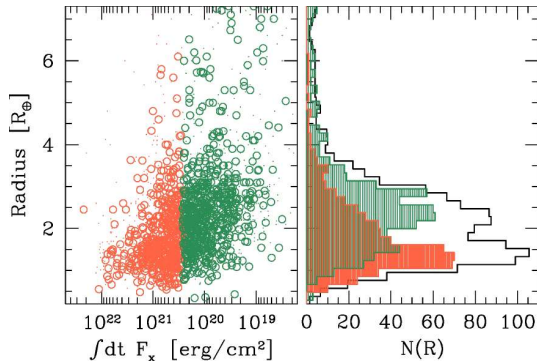
$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

- Young stars saturate XUV flux at $\sim 10^3 \times$ modern levels for ~ 100 Myr
- Strips light primordial H/He envelopes from lower-mass cores at close-in orbits
- Owen & Wu (2013): naturally carves the bimodal radius valley

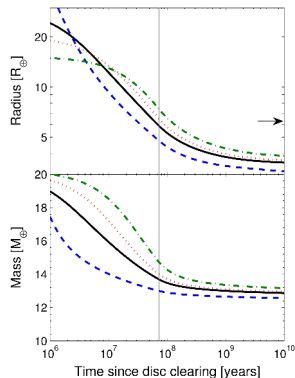


Core-powered mass loss model of the radius valley: envelope loss driven by heat from the cooling rocky interior produces a bimodal distribution with a gap at $\sim 1.8 R_{\oplus}$. Credit: Ginzburg et al. (2018).

Photoevaporation Model + Mass-Loss Tracks

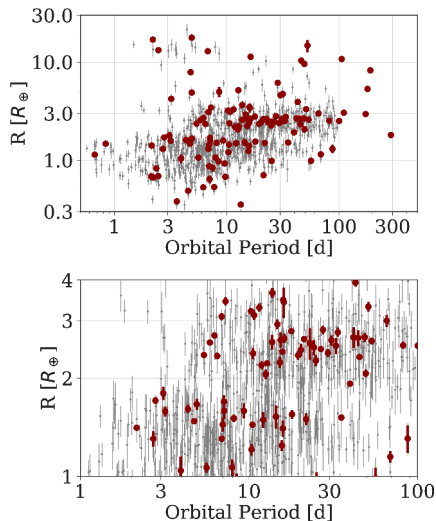


Left: predicted radius valley



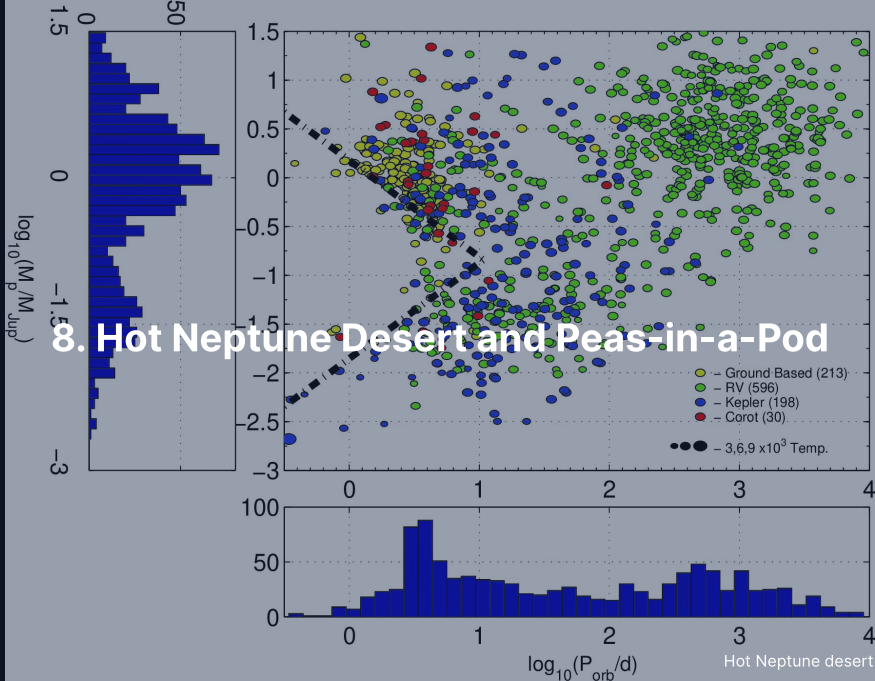
Right: time evolution of radius and mass; saturated XUV phase drives bimodal outcome (Owen & Wu 2013).

Valley Slope and Stripped Cores

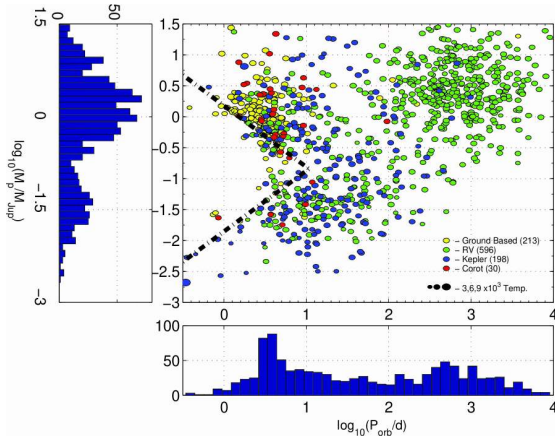


Credit: Van Eylen et al. (2018)

- Asteroseismic radii reveal the valley slope shifts to smaller R_p at longer periods
- Consistent with photoevaporation and core-powered cooling models
- Many close-in super-Earths are stripped cores rather than primordial rocky worlds



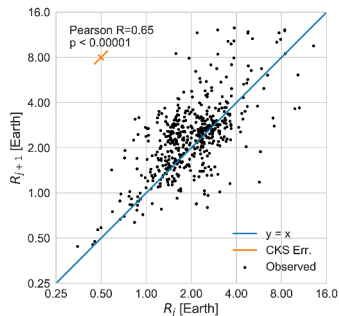
The Hot Neptune Desert



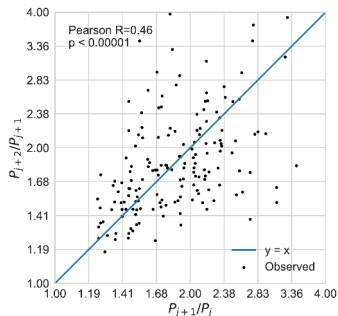
Credit: Mazeh et al. (2016)

- Pronounced deficit of short-period planets ($P < 5 \text{ d}$) in the $10\text{--}100 M_{\oplus}$ mass range
- **Bottom edge:** Photoevaporation strips low-mass envelopes into super-Earths
- **Top edge:** Tidal Roche-lobe overflow strips inflated gas giants

Peas-in-a-Pod Architectures



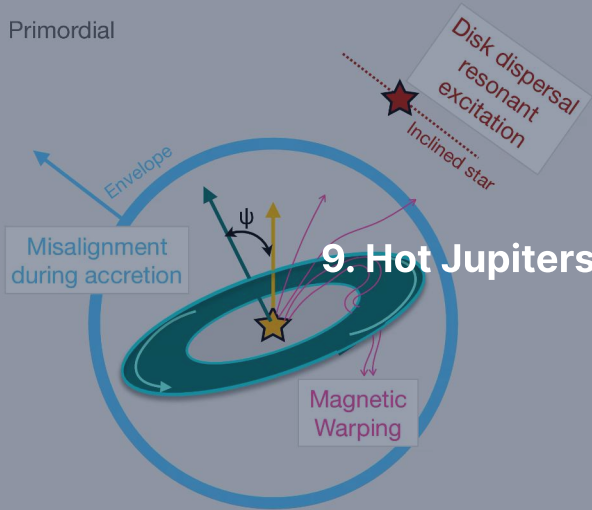
Adjacent radii: Pearson $r = 0.65$



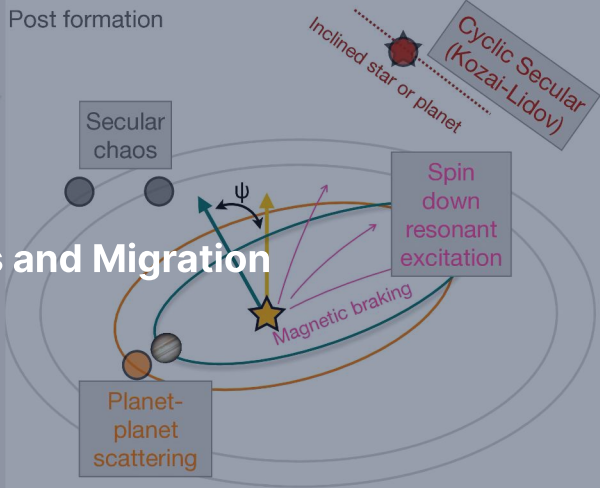
Adjacent period ratios: clustered

- Planets in compact multi-systems tend to have similar sizes and uniform orbital spacing (Weiss et al. 2018)
- Favours quiet in-situ growth or migration over chaotic giant impacts

Primordial

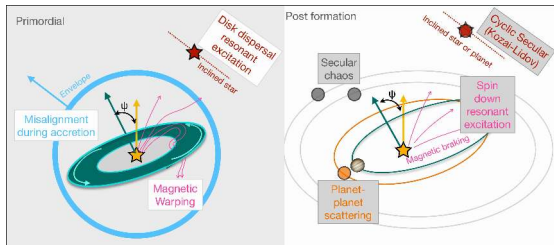


Post formation



9. Hot Jupiters and Migration

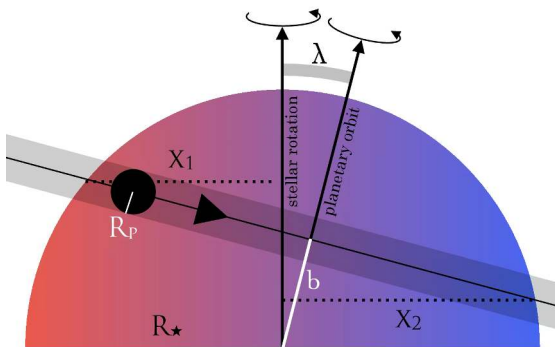
Hot Jupiter Migration Channels



Credit: Albrecht et al. (2022)

- **Disk migration (Type II):** disk torques move the planet inward smoothly, preserving low obliquity (in situ growth at 0.05 AU fails: too little solid mass, too hot)
- **High-eccentricity (Kozai-Lidov):** distant perturbers pump e ; tidal circularisation leaves wide misalignments
- **Planet scattering:** instability ejects giants and scrambles alignments

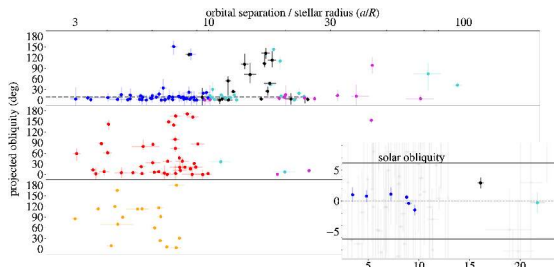
Rossiter-McLaughlin Effect



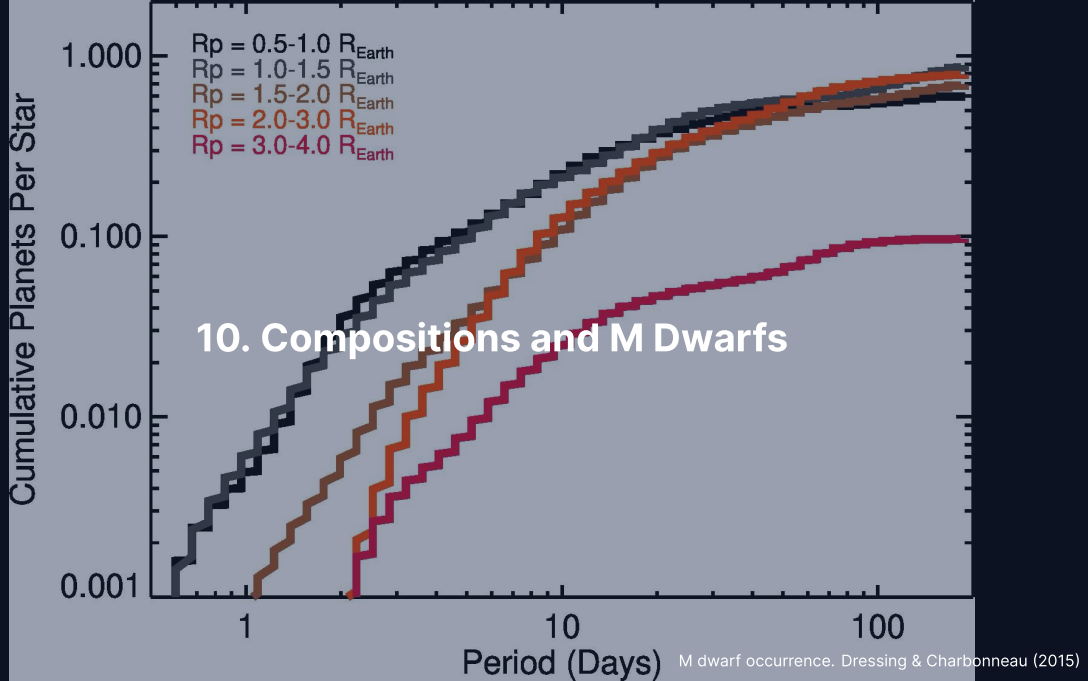
- Transiting planet first occults blueshifted (approaching) hemisphere, then redshifted
- Time-resolved line distortion traces sky-projected spin-orbit angle λ
- Diagnostic tool distinguishing disk migration from dynamical scattering (Triaud 2018)

Observed Obliquity Distribution

Stellar obliquity distributions show that cool stars realign tidally while hot stars retain widely scattered spin-orbit orientations.



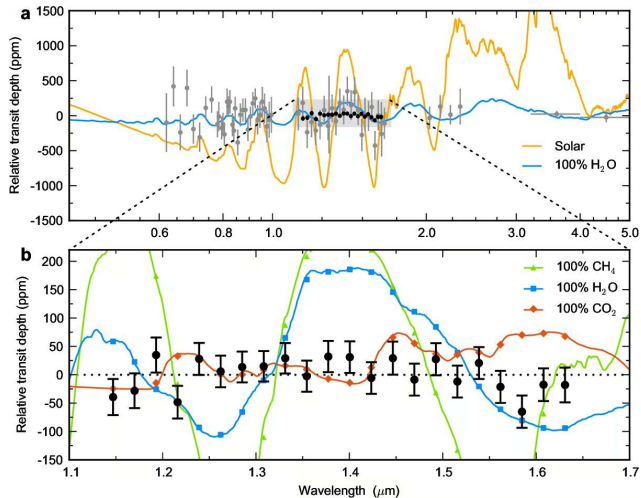
- **Cool stars ($T_{\text{eff}} < 6250 \text{ K}$):** Tight orbits aligned by tidal realignment in convective envelopes
- **Hot stars ($T_{\text{eff}} > 6250 \text{ K}$):** Retain high obliquities due to thin convective zones
- **Primordial scatter:** Wide dispersion at large $a/R_{\star}$ proves dynamical migration occurs



Sub-Neptunes: Water World or H/He?

Sub-Neptune bulk densities suffer a severe degeneracy between extended hydrogen-helium envelopes and deep volatile water layers.

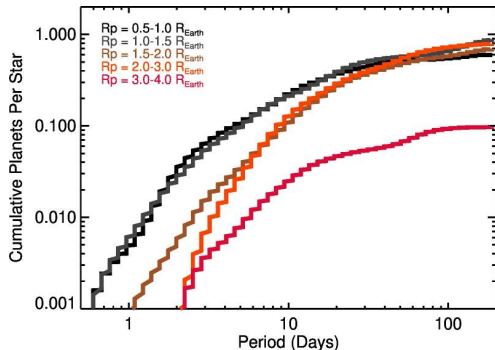
- **Density split:** Super-Earths ($\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$) are rocky; sub-Neptunes ($\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$) require volatile envelopes
- **Degeneracy:** Density cannot separate a 1% H/He envelope from a 50% water-rich mantle
- **Spectroscopic test:** Transmission spectroscopy is required to determine the atmospheric mean molecular weight



Featureless transmission spectrum of the warm sub-Neptune GJ 1214 b from HST data: ruling out cloud-free H_2O , CH_4 , and CO_2 atmospheres demonstrates high-altitude clouds or hazes.

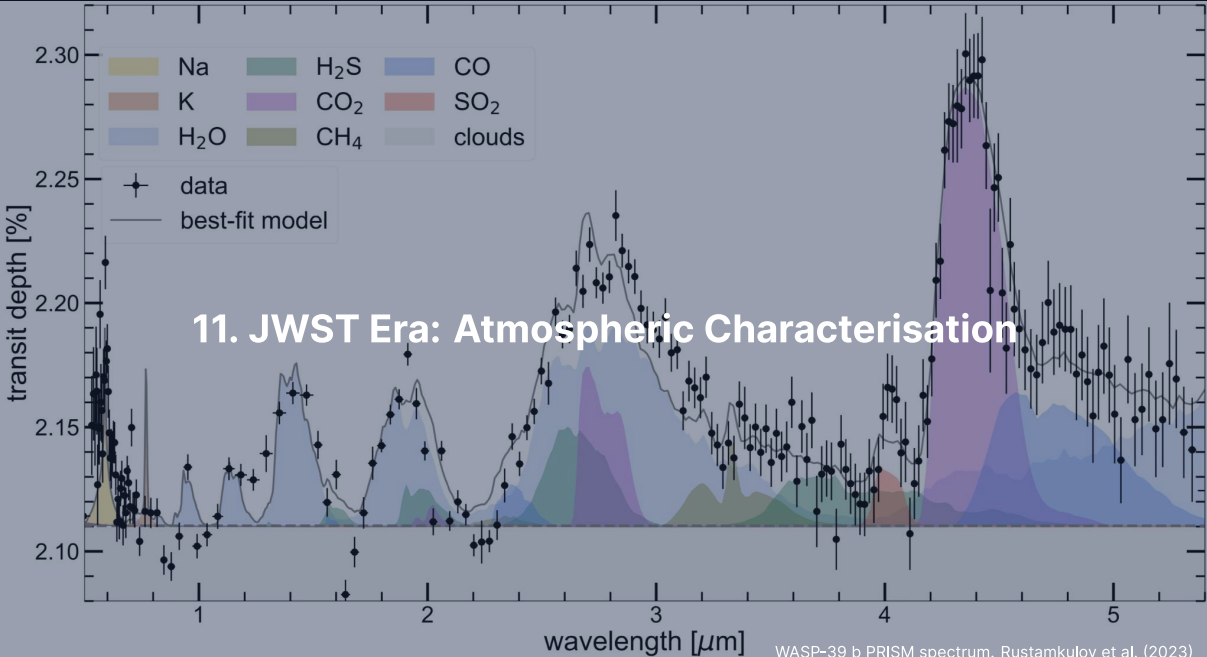
Credit: Kreidberg et al. (2014).

M Dwarfs: Abundant Hosts, Severe Hazards



Credit: Dressing & Charbonneau (2015)

- M dwarfs host ~ 2.5 small planets each; geometrically favourable for transits and RV
- **XUV desiccation:** Extended pre-main-sequence luminosity can strip entire oceans (Luger & Barnes 2015)
- **Tidal locking:** 1:1 spin-orbit resonance forces permanent day and night sides



Transmission Spectroscopy

Starlight filtered through an atmosphere during transit encodes molecular absorption.

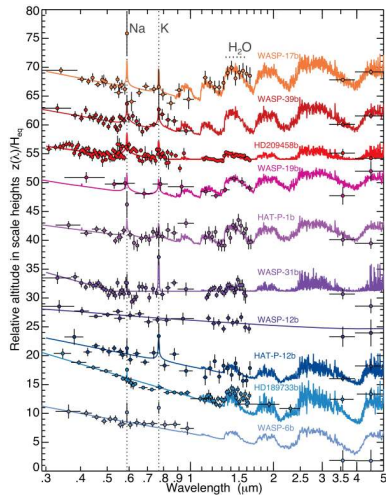
Atmospheric scale height:

$$H = \frac{k_B T}{\mu m_u g}$$

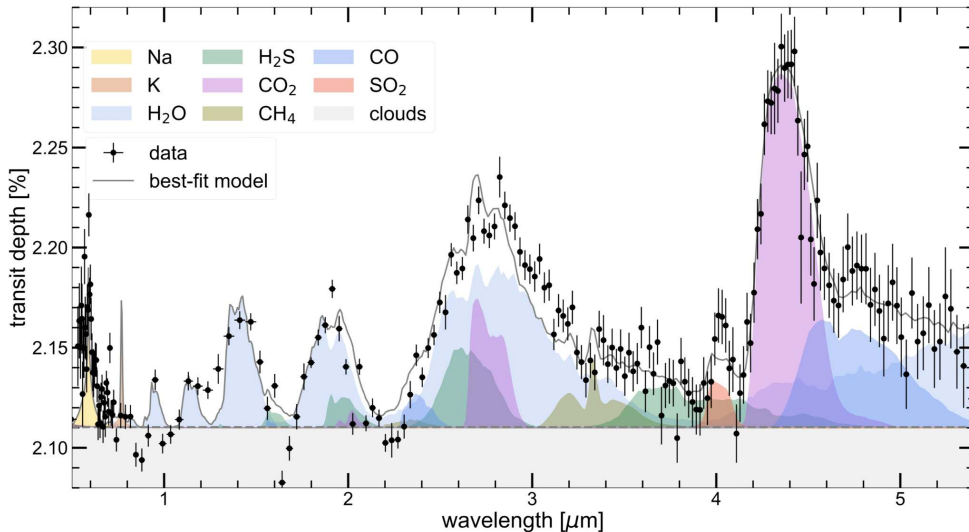
Wavelength-dependent transit depth: $\delta(\lambda) \approx (R_p + n_H(\lambda) H)^2 / R_\star^2$, so $\Delta\delta/\delta \approx 2n_H H / R_p$.

For a hot Jupiter ($T = 1500$ K, $\mu = 2.3$, $g = 25 \text{ m s}^{-2}$):

- Scale height $H \approx 200$ km
- Absolute spectral modulation: ~ 200 ppm (JWST-routine)
- Sub-Neptunes: 10–100 ppm; terrestrial atmospheres: < 10 ppm
- High clouds or hazes flatten the spectrum (GJ 1214 b): a flat spectrum does not mean no atmosphere

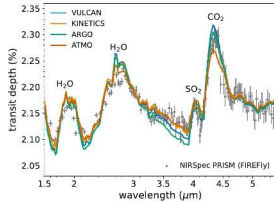


Transmission spectra of ten hot Jupiters from HST and Spitzer across 0.3–5 μm , revealing a continuum from clear atmospheres with prominent water absorption to cloudy, aerosol-dominated scattering slopes. Credit: Sing et al. (2016).

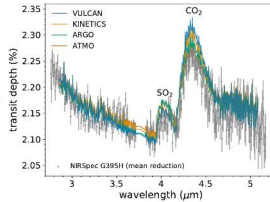


JWST/NIRSpec PRISM transmission spectrum of WASP-39 b. Shaded bands are model opacity contributions; data demonstrate detections of H₂O (33σ), CO₂ (28σ), clouds (21σ), Na (19σ), and CO (7σ). Rustamkulov et al. (2023).

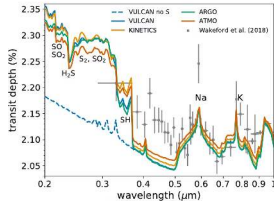
WASP-39 b: First Disequilibrium Photochemistry



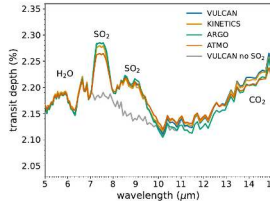
(a)



(b)

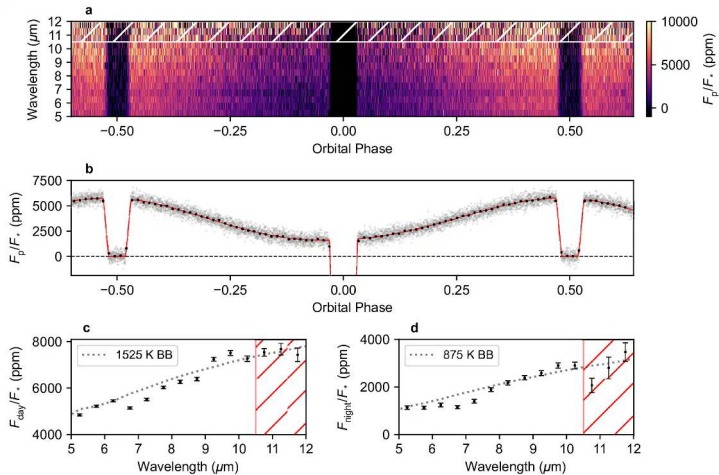


(c)



(d)

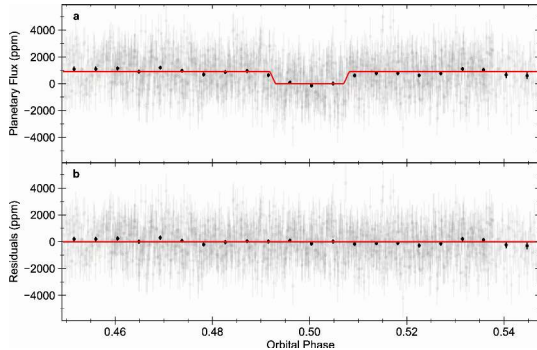
- Hot Saturn ($0.28 M_J$) early release science target
- $4.05 \mu\text{m}$ absorption feature confirms **photochemical** SO_2
- Requires UV photolysis of H_2S , proving disequilibrium chemistry
- First photochemical product identified in an exoplanet atmosphere (Tsai et al. 2023)



JWST/MIRI spectroscopic phase curve of WASP-43 b across orbital phase, showing thermal emission variation from dayside to nightside and constrained atmospheric heat redistribution.

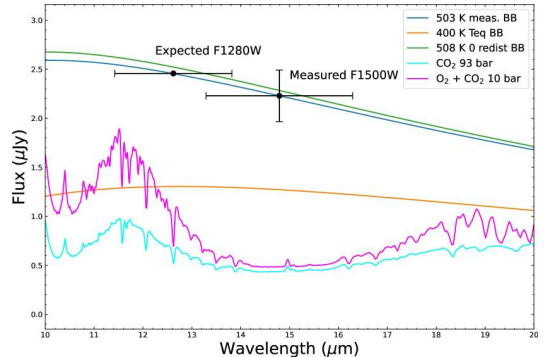
Credit: Bell et al. (2024).

TRAPPIST-1 b: Bare Rock Dayside

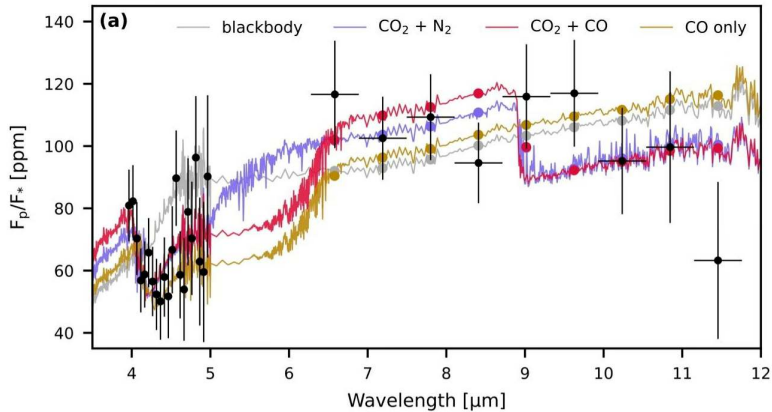


15 μm secondary eclipse

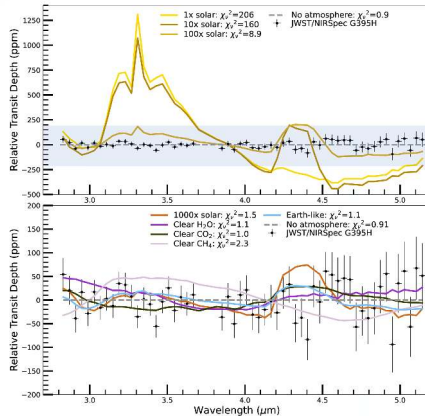
$T_B = 503^{+26}_{-27}$ K, consistent with bare rock equilibrium. **Thick CO₂ atmosphere ruled out**; inner M dwarf rocky planets show a widespread lack of thick atmospheres. Greene et al. (2023).



Compared to model atmospheres

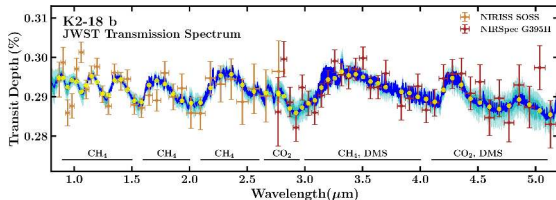


JWST NIRCam and MIRI thermal emission spectrum of 55 Cancri e: the binned data fall below the bare-rock blackbody at most wavelengths and favour atmospheric models with CO and CO_2 , tentative evidence for a secondary atmosphere outgassed from a molten surface.



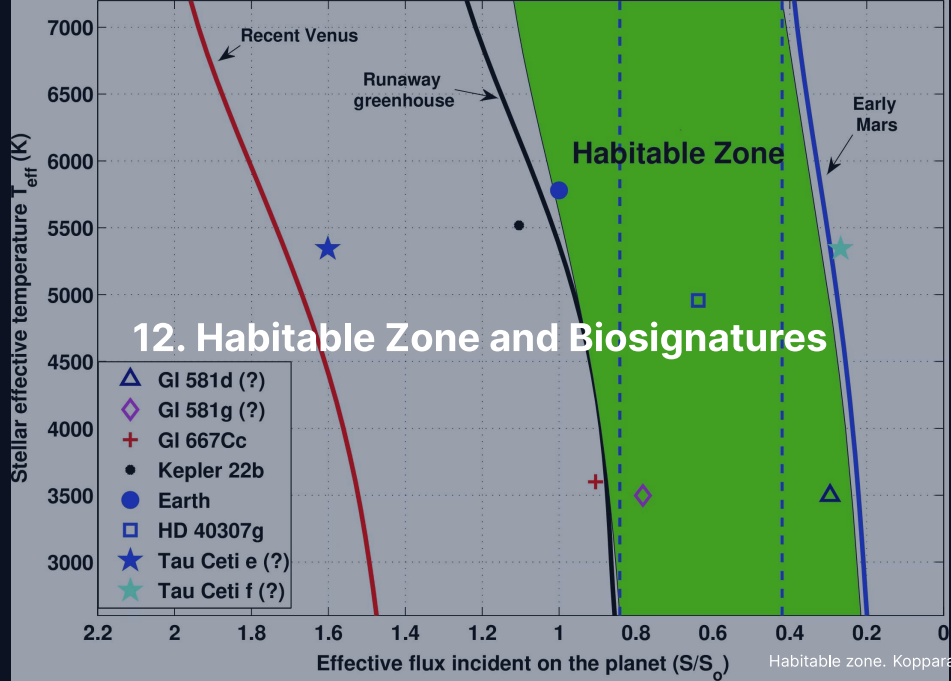
JWST NIRSpec G395H transmission spectrum of LHS 475 b, an Earth-size M dwarf planet at about 12 pc: flat and featureless. Hydrogen-helium atmospheres are ruled out at high confidence, a pure CH₄ atmosphere too, and a Venus-like pure CO₂ atmosphere remains marginally consistent.

K2-18 b: Testing a Tentative Biosignature

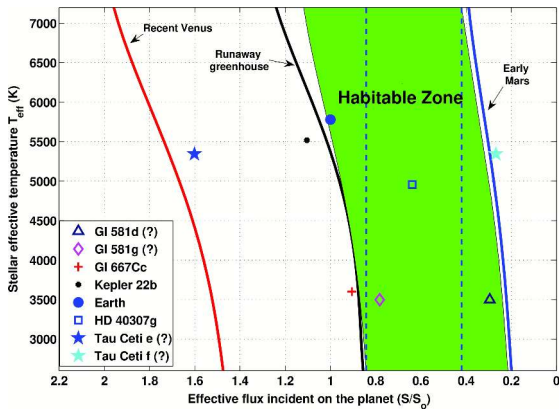


Credit: Madhusudhan et al. (2023)

- Sub-Neptune in M-dwarf HZ with detected CH_4 and CO_2 ; disputed tentative DMS ($\lesssim 2\sigma$)
- Competing models: Hycean ocean, mini-Neptune envelope, or magma ocean
- Exemplifies rigorous real-time vetting required for candidate biosignatures



The Classical Habitable Zone

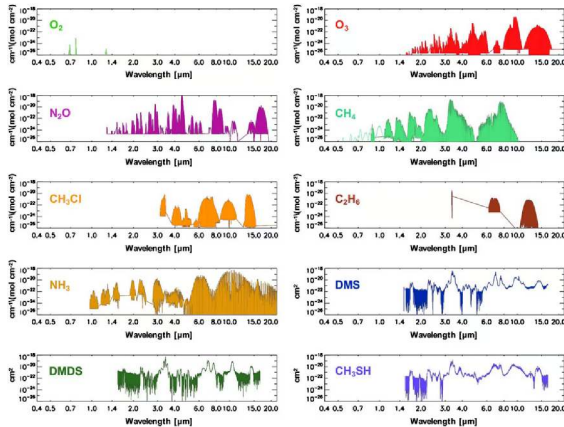


- Inner edge: Simpson-Nakajima runaway greenhouse threshold
- Outer edge: Maximum CO_2 greenhouse before CO_2 condenses
- Boundaries scale with stellar effective temperature and luminosity
- Updated 1D radiative-convective grid from Kopparapu et al. (2013)

Habitable Zone Evolution and Biosignatures

- **Dynamic trajectory:** HZ boundaries shift with stellar evolution; pre-MS M dwarfs subject planets to prolonged runaway desiccation
- **Disequilibrium pairs:** Atmospheric biosignatures demand chemical disequilibrium ($\text{O}_2 + \text{CH}_4$), not lone gas species
- **False positives:** Abiotic O_2 can build up via H_2O photolysis with hydrogen escape or CO_2 photolysis

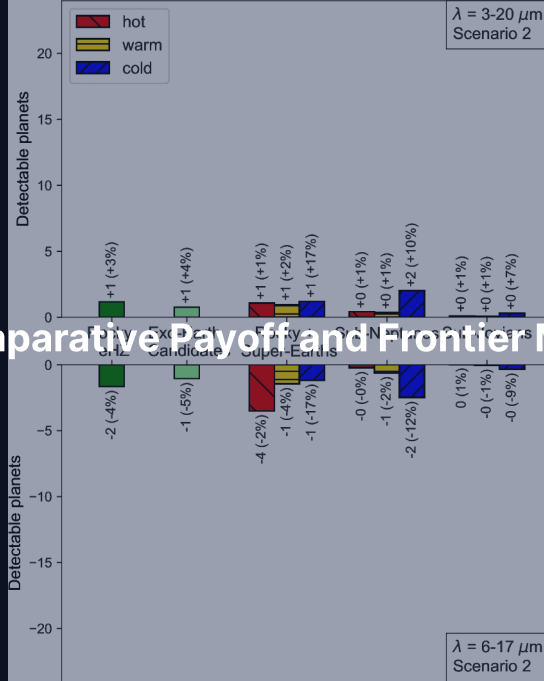
Life detection is an inverse problem requiring exclusion of all abiotic pathways.

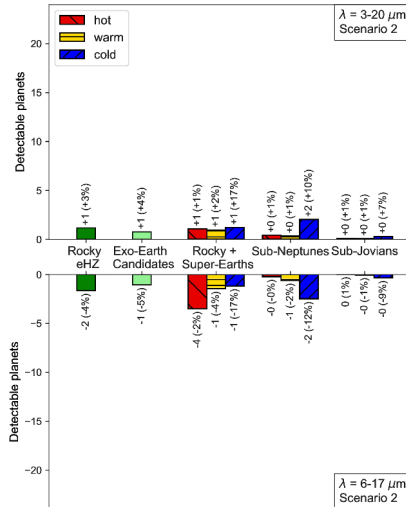


Absorption cross-sections from 0.4 to 20 μm for ten candidate biosignature gases (O₂, O₃, N₂O, CH₄, CH₃Cl, C₂H₆, NH₃, dimethyl sulfide, dimethyl disulfide, CH₃SH): oxygen absorbs only in narrow optical bands, the others have their strongest features in the infrared.

Reproduced from Schwieterman et al. (2018).

13. Comparative Payoff and Frontier Missions





Predicted detection yields of the LIFE mid-infrared space interferometer for exoplanets within 20 pc, by radius and insolation. Quanz et al. (2022).

Summary: Today's Course-Level Takeaways

1. **Detection biases:** Transit plus radial velocity breaks $m \sin i$ to yield true bulk density
2. **Demographics:** Small planets dominate ($\eta_{\oplus} \sim 0.4$); photoevaporation carves the radius valley at $\sim 1.8 R_{\oplus}$
3. **Architectures:** resonant chains and peas-in-a-pod systems record smooth disk migration and a shared formation environment
4. **Characterisation:** JWST detects photochemistry and bare rocks; biosignatures require ruling out false positives

Next: Lecture 14



Questions?

Next: Lecture 14. Synthesis & Astrobiology

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience